

WAINGAPU MAU HAU, Indonesia

WMO: 973400

Lat: 9.6697S

Lon: 120.2997E

Elev: 39

StdP: 14.68

Time Zone: 8.00 (E08)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	67.2	68.4	58.8	74.0	77.8	61.0	80.2	76.0	23.4	85.2	20.5	86.4	5.7	220	0.346

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	13.8	92.4	79.0	91.6	79.3	90.8	79.1	82.5	89.6	81.7	89.0	81.0	88.5	12.5	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.4	157.9	88.0	79.5	153.2	87.3	78.8	149.8	86.8	46.3	89.7	45.5	89.4	44.8	88.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.7	17.3	15.3	DB	64.3	95.4	2.0	1.4	62.9	96.4	61.7	97.3	60.5	98.1	59.1	99.1	(4)
(5)			WB	60.1	84.5	1.5	2.2	59.0	86.1	58.1	87.3	57.3	88.6	56.2	90.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.3	82.5	82.0	82.4	82.1	81.2	79.5	78.5	78.4	80.0	82.4	83.6	83.4
	DBStd	2.44	1.68	1.52	1.43	1.41	1.77	2.25	1.92	1.92	1.95	1.62	1.30	1.64
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11435	1007	895	1006	963	966	885	884	882	900	1004	1009	1034
	CDD65	5960	542	475	541	513	501	435	419	417	450	539	559	569
	CDH74	61196	5643	4685	5562	5391	4983	3891	3557	3703	4533	6245	6678	6325
	CDH80	23025	1939	1539	1975	2070	1853	1313	1214	1368	1795	2704	2842	2413
Wind	WSAvg	5.9	4.3	4.5	5.1	5.4	6.2	6.4	7.0	7.1	7.0	6.7	6.1	5.0
Precipitation	PrecAvg	35.0	7.0	7.1	6.6	2.8	1.7	0.7	0.4	0.2	0.0	0.8	1.8	5.7
	PrecMax	56.6	13.3	11.2	13.2	9.2	7.3	8.6	2.8	1.3	1.0	6.3	5.6	10.9
	PrecMin	18.9	1.8	2.7	1.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.3
	PrecStd	9.7	2.8	2.5	2.8	2.4	2.0	1.7	0.7	0.4	0.2	1.4	1.8	2.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.6	91.4	91.8	91.8	90.9	89.8	89.2	90.0	91.4	93.3	93.4	92.8
		MCWB	78.4	80.0	80.4	79.1	77.9	76.7	74.6	74.3	76.1	76.5	79.0	80.4
	2%	DB	90.8	90.1	90.6	90.7	89.8	88.6	87.9	88.6	90.0	91.8	92.1	91.7
		MCWB	79.3	79.7	80.4	79.2	77.6	76.5	74.7	74.1	76.3	77.6	79.5	80.7
	5%	DB	89.4	89.0	89.6	89.7	88.8	87.5	86.9	87.6	89.0	90.7	91.2	90.7
		MCWB	79.5	79.5	80.2	79.0	77.5	76.1	74.3	74.2	76.3	77.9	79.8	80.5
	10%	DB	88.0	87.7	88.3	88.7	87.7	86.4	85.9	86.4	87.8	89.7	90.2	89.2
		MCWB	79.5	79.4	79.8	78.9	77.4	75.7	73.9	74.0	76.3	77.9	79.7	80.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	83.0	82.1	82.6	82.1	81.5	80.0	78.8	79.0	80.3	81.7	83.1	83.8
		MCDB	88.8	88.6	90.0	89.1	88.5	87.0	85.6	86.9	88.5	89.6	90.5	90.2
	2%	WB	81.5	81.0	81.5	81.2	80.4	78.6	77.4	77.3	79.1	80.6	81.9	82.4
		MCDB	88.2	88.0	89.0	88.4	87.3	86.0	85.1	86.1	87.5	89.3	90.0	89.6
	5%	WB	80.5	80.3	80.7	80.5	79.5	77.7	76.4	76.3	78.2	79.7	81.0	81.2
		MCDB	87.6	87.4	88.2	87.9	86.2	85.1	84.3	85.4	86.6	88.5	89.3	88.6
	10%	WB	79.7	79.6	80.0	79.8	78.6	76.7	75.4	75.3	77.3	78.9	80.3	80.4
		MCDB	86.4	86.3	87.1	86.9	85.2	84.0	83.3	84.5	85.8	87.8	88.6	87.8
Mean Daily Temperature Range	5% DB	MDBR	11.8	11.9	12.6	13.4	13.7	14.4	15.9	16.9	16.2	15.4	13.8	12.4
		MCDBR	13.5	13.4	13.7	14.5	14.9	14.7	16.5	17.6	17.1	16.5	14.9	13.9
		MCWBR	5.7	6.1	6.6	6.9	6.5	6.2	7.3	8.1	8.5	8.2	7.4	6.3
	5% WB	MCDBR	12.2	12.5	13.1	13.1	13.0	13.5	14.8	16.5	15.5	14.9	14.0	13.0
		MCWBR	6.4	6.4	6.8	7.1	7.0	6.9	7.7	9.1	8.5	8.1	7.5	6.8
Clear-Sky Solar Irradiance	taub	0.435	0.430	0.417	0.416	0.424	0.417	0.406	0.410	0.435	0.469	0.469	0.451	
	taud	2.408	2.423	2.452	2.456	2.424	2.435	2.455	2.422	2.347	2.233	2.270	2.347	
	Ebn at Noon	288	288	287	278	266	263	269	276	278	274	276	283	
	Edh at Noon	40	39	37	36	35	34	34	36	41	47	45	42	
All-Sky Solar Radiation	RadAvg	1780	1840	1846	1810	1677	1557	1643	1868	2073	2167	2079	1853	
	RadStd	183	181	154	128	93	81	44	58	93	85	112	126	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page